

Johnny Rhe

SungKyunKwan University: 2066, Seobu-ro, Jangan-gu, Suwon-si, Gyeonggi-do, Korea

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Research Interests

I am currently pursuing a Ph.D. degree in the Department of Electrical and Computer Engineering at Sungkyunkwan University. My research focuses on algorithm-hardware co-design for efficient In-Memory Computing (IMC)-based Deep/Convolutional Neural Networks (D/CNNs) inference. I have worked on a wide range of topics including mapping optimization via pruning and compression techniques, reliability enhancement on analog IMC arrays, and a software simulation framework. Recently, I have expanded my research interests to transformer-based models like Vision Transformer (ViT) and Large Language Models (LLMs), aiming to optimize their performance and inference efficiency in IMC systems. To achieve this, I am exploring various hardware-aware optimization techniques, with the ultimate goal of enabling scalable and energy-efficient deployment of these models on IMC systems.

Education

Ph.D. Electrical and Computer Engineering

Sept. 2019 – Present

Sungkyunkwan University (SKKU), Suwon, Korea

- GPA: 3.8/4.5
- Intelligent & Resource-Efficient Image Processing & Systems Design (IRIS) Lab
- Thesis Advisor: Prof. [Jong Hwan Ko](#)

B.S. School of Electronic and Electrical Engineering

Mar. 2011 - Feb. 2019

SungKyunKwan University (SKKU), Suwon, Korea

- GPA: 3.8/4.5 (Top 22%)

Research Experience

Accelerated Computation Algorithms for IMC-based D/CNN Inference

Sept. 2019 - Present

- Developed novel DNN weight mapping methods [[C2](#), [C15](#)]
- Proposed multi-level mapping optimization via pruning [[J2](#), [C4](#), [J5](#), [C5](#), [J3](#)]
- Proposed low-rank and kernel grouping methods for compact weight representation [[C11](#), [C7](#)]
- Proposed input-stationary mapping methods for energy-efficient and flexible inference [[C3](#), [J4](#)]

Reliability Enhancement on Analog IMC Arrays

Sept. 2022 - 2024

- Designed fault-robust remapping schemes to mitigate inference error [[C5](#)]
- Proposed hybrid grouping to improve multi-bit weight precision and fault tolerance [[C12](#)]

Software-based Simulator for IMC-based Inference

Sept. 2021 - 2023

- Designed visualization and evaluation frameworks integrated with NeuroSim [[C9](#)]

Resource-efficient Neural Networks for Edge Applications

Sept. 2023 - 2024

- Proposed lightweight CNN/SNN models for biomedical signal decoding [[C13](#), [C8](#)]

Projects

A Study on Collaborative Mapping and Training Techniques for Energy-Efficient In-Memory CNN Inference

Sept. 2024 - Present

National Research Foundation (NRF), Korea

-  [[G1](#)] Project Leader

Optimizing DNN Mapping and Execution on In-Memory Computing Arrays

April. 2024 - Present

Ministry of Trade, Industry & Energy (MOTIE), Korea

HW-SW Co-Design for Low-Power Real-Time In-Memory Deep Learning Operation

Mar. 2023 - Present

Samsung Advanced Institute of Technology (SAIT), Korea

Spatiotemporal Precision Control for Efficient and Robust In-Memory Deep Learning Inference

June. 2023 - May. 2024

National Research Foundation (NRF), Korea

Honors & Awards

[H16] 2 nd /3 rd Place Winners at Grand Challenge on Brain-Machine Interfaces <i>IEEE Biomedical Circuits and Systems Conference (BioCAS)</i>	Oct, 2025
[H15] 2025 SKKU Innovative Research Fellowship <i>Sungkyunkwan University (SKKU)</i>	May, 2025
[H14] Poster Presentation at PhD Forum <i>Design, Automation and Test in Europe Conference (DATE) 2025</i>	Mar, 2025
[H13] Outstanding Research Award <i>Korea Collaborative & High-tech Initiative for Prospective Semiconductor Research (K-CHIPS)</i>	Nov, 2024
[H12] KETEP Outstanding Research Award <i>Korea Institute of Energy Technology Evaluation and Planning (KETEP)</i>	Oct, 2024
[H11] Samsung Outstanding Paper Award - Samsung Elec. & SKKU (2024) <i>Samsung Electronics</i>	Aug, 2024
[H10] Grand Prize at the Exynos AI Challenger <i>Samsung Electronics' S.LSI</i>	July, 2024
[H9] 3 rd Place Winner in the System Design Contest (GPU Track) <i>Design Automation Conference (DAC) 2024</i>	June, 2024
[H8] Poster Presentation at PhD Forum <i>IEEE International Symposium on Circuits and Systems (ISCAS) 2024</i>	May, 2024
[H7] Most Popular Poster Award at the Student Research Forum <i>29th Asia South Pacific Design Automation Conference (ASP-DAC) 2024</i>	Jan, 2024
[H6] Poster Presentation at Student Research Forum <i>29th Asia South Pacific Design Automation Conference (ASP-DAC) 2024</i>	Jan, 2024
[H5] Samsung Best Paper Award - Samsung Elec. & SKKU (2023) <i>Samsung Electronics</i>	Aug, 2023
[H4] IEEE CASS Non-Flagship Conference Student Travel Grant (Selection) - ISLPED <i>IEEE Circuits and Systems Society (CAS)</i>	Aug, 2023
[H3] Best Paper Award <i>Korean Artificial Intelligence Association (KAIA)</i>	July, 2023
[H2] 2022 SKKU Innovative Research Fellowship <i>Sungkyunkwan University (SKKU)</i>	Sept, 2022
[H1] 2 nd Place Winner of Artificial Intelligence Grand Challenge <i>Institute of Information & Communications Technology Planning & Evaluation, Korea</i>	July, 2020

Featured Publications

JOURNAL ARTICLES

[J5] **Johnny Rhe**, Kang Eun Jeon and Jong Hwan Ko, "Genetic Algorithm-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping," *Journal of Systems Architecture (JSA)*, May. 2025 (JCR Q1). [[Paper](#)]

[J3] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, "KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping," *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)* vol. 71, no. 12, pp. 6138–6151, Nov. 2024 (JCR Q1). [[Paper](#)]

🏆 [H12] Outstanding Research Award from KETEP (2024)

[J2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)* vol. 12, no. 2, pp. 408–421, May. 2022 (JCR Q1). [\[Paper\]](#)

CONFERENCE PROCEEDINGS

[C15] **Johnny Rhe**, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, “ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures,” *Asia Pacific Conference on Circuits and Systems (APCCAS)* (Accepted), Oct. 2025.

[C5] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2023. [\[Paper\]](#)

🏆 [\[H11\]](#) Outstanding Paper Award from Samsung Elec.(2024)

[C4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, “PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures,” *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [\[Paper\]](#)

🏆 [\[H4\]](#) Student Travel Grant Selection from IEEE CAS

[C2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures,” *Design, Automation, and Test in Europe (DATE)*, Mar. 2022 (Oral, acceptance rate: 25%). [\[Paper\]](#)

Full Publications (110+ Citations)

Most-up-to-date publication list can be founded at [Google Scholar](#)

JOURNAL ARTICLES

[J5] **Johnny Rhe**, Kang Eun Jeon and Jong Hwan Ko, “Genetic Algorithm-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping,” *Journal of Systems Architecture (JSA)*, May. 2025 (JCR Q1). [[Paper](#)]

[J4] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, “Input/Mapping Precision Controllable Digital PIM with Adaptive Adder Tree Architecture for Flexible DNN Inference,” *Journal of Systems Architecture (JSA)*, Feb. 2025 (JCR Q1). [[Paper](#)]

[J3] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)* vol. 71, no. 12, pp. 6138–6151, Nov. 2024 (JCR Q1). [[Paper](#)]

🏆 [H12] Outstanding Research Award from KETEP (2024)

[J2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)* vol. 12, no. 2, pp. 408–421, May. 2022 (JCR Q1). [[Paper](#)]

[J1] Eunyong Lee, Taeyoung Han, Donguk Seo, Gicheol Shin, Jaerok Kim, Seonho Kim, Soyoun Jeong, **Johnny Rhe**, Jaeyung Park, Jong Hwan Ko, Yoonmyung Lee, “A Charge-Domain Scalable-Weight In-Memory Computing Macro with Dual-SRAM Architecture for Precision-Scalable DNN Accelerators,” *IEEE Transactions on Circuits and Systems I (TCAS-I)* vol. 68, no. 8, pp. 3305–3316, May. 2021 (JCR Q1). [[Paper](#)]

CONFERENCE PROCEEDINGS

[C15] **Johnny Rhe**, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, “ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures,” *Asia Pacific Conference on Circuits and Systems (APCCAS)* (Accepted), Oct. 2025.

[C14] **Johnny Rhe**, and Jong Hwan Ko, “Weight Sharing for Array-Efficient CNN Inference in Compute-In-Memory Architectures,” *International SoC Design Conference (ISOCC)* (Accepted), Oct. 2025.

[C13] Chanwook Hwang, Chan Woong Park, **Johnny Rhe**, and Jong Hwan Ko, “Efficient and Robust SNN Decoder Training for Closed-Loop Brain-Machine Interfaces,” *IEEE Biomedical Circuits and Systems Conference (BioCAS)* (Accepted), Oct. 2025.

🏆 [H16] 2nd/3rd Place Winner at Grand Challenge by BioCAS

[C12] Kang Eun Jeon, Sangheum Yeon, Jinhee Kin, Hyeonsu Bang, **Johnny Rhe**, and Jong Hwan Ko, “Row-Column Hybrid Grouping for Fault-Resilient Multi-Bit Weight Representation on IMC Arrays,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2025. [[Paper](#)]

[C11] Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, “Low-Rank Compression for IMC Arrays,” *Design, Automation & Test in Europe Conference (DATE)*, Mar. 2025. [[Paper](#)]

[C10] **Johnny Rhe**, and Jong Hwan Ko, “Row-Efficient Pruning for In-Memory Convolutional Weight Mapping,” *International SoC Design Conference (ISOCC)*, Aug. 2024. [[Paper](#)]

[C9] Kang Eun Jeon¹, Wooram Seo¹, **Johnny Rhe**, and Jong Hwan Ko, “ConvMapSim: Modeling and Simulating Convolutional Network Mapping on PIM Arrays,” *IEEE Artificial Intelligence Circuits and Systems (AICAS)*, Apr. 2024. [[Paper](#)]

[C8] Chanwook Hwang, Jaehyeon So, **Johnny Rhe**, Jiyeon Kim, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, “An Efficient Ventricular Arrhythmias Detection on Microcontrollers with Optimized 1D CNN,” *IEEE Artificial Intelligence Circuits and Systems (AICAS)*, Apr. 2024. [[Paper](#)]

🏆 [H10] Grand Prize at Exynos AI Challenger by Samsung Elec.

[C7] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, “KARS: Kernel-Grouping Aided Row-Skipping for SDK-based Weight Compression in PIM Arrays,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2024. [[Paper](#)]

🏆 [H13] Best Paper Award from KAIA (2023)

[C6] Hyeonsu Bang, Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, “DCR: Decomposition-Aware Column Re-Mapping for Stuck-At-Fault Tolerance in ReRAM Arrays,” *IEEE International Conference on Computer Design (ICCD)*, Nov. 2023. [[Paper](#)]

[C5] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, “Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays,” *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2023. [[Paper](#)]

🏆 [H11] Outstanding Paper Award from Samsung Elec. (2024)

[C4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, “PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures,” *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [[Paper](#)]

🏆 [H4] Student Travel Grant Selection from IEEE CAS

[C3] Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, “Weight-Aware Activation Mapping for Energy-Efficient Convolution on PIM Arrays,” *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [[Paper](#)]

🏆 [H5] Outstanding Paper Award from Samsung Elec. (2023)

[C2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures,” *Design, Automation, and Test in Europe (DATE)*, Mar. 2022 (Oral, acceptance rate: 25%). [\[Paper\]](#)

[C1] Gicheol Shin, Donguk Seo, Jaerok Kim, **Johnny Rhe**, Eunjung Lee, Seonho Kim, Soyoun Jeong, Jong Hwan Ko, Yoonmyung Lee, “A charge-Domain Computation-in-Memory Macro with Versatile All-Around-Wire-Capacitor for Variable-Precision Computation and Array-Embedded DA/AD Conversions,” *European Conference on Solid-State Circuits (ESSCIRC)*, Sept. 2021. [\[Paper\]](#)

Patents

[P9] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, US Patent Application No.19069410, method and electronic device with weight pruning cross-reference to related application, Mar. 2025

[P8] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, and Jong Hwan Ko, PCT Patent Application No.PC-T/KR2024/016234, Method and apparatus for convolution operation utilizing kernel shape control, Oct. 2024

[P7] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, KR Patent Application No.10-2024-0058360, Pattern-based weight pruning method and electronic devices, May. 2024

[P6] **Johnny Rhe**, and Jong Hwan Ko, US Patent Application No.18398389, Apparatus and method for controlling process-in-memory by accelerating convolution operation based on arranging pattern of weight in kernel, and storage medium storing instructions to perform method for controlling process-in-memory, Dec. 2023

[P5] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, KR Patent Application No.10-2023-0157705, Adaptive adder tree and digital in-memory architecture supporting multi-bit partitioning and improved column-wise mapping for depth-wise convolution layers, Nov. 2023

[P4] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, and Jong Hwan Ko, KR Patent Application No.10-2023-0171519, Method and apparatus for convolution operation utilizing kernel shape control, Nov. 2023

[P3] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, US Patent Application No.18090628, Memory device for optimizing computation of convolution layer, method for controlling memory device, and recording medium storing instruction to perform method for controlling memory device, Dec. 2022

[P2] **Johnny Rhe**, and Jong Hwan Ko, KR Patent Application No.10-2022-0187676, Apparatus and method for controlling process-in-memory by accelerating convolution operation based on arranging pattern of weight in kernel, and storage medium storing instructions to perform method for controlling process-in-memory, Dec. 2022

[P1] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, KR Patent No.No.10-2021-0191289, Memory device for optimizing computation of convolution layer, method for controlling memory device, and recording medium storing instruction to perform method for controlling memory device, Dec. 2021

Talks

Optimizing Weight Mapping for Energy-Efficient In-Memory CNN Inference Oct, 2025
International Conference on Computer-Aided Design (ICCAD)

ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures Oct, 2025
Asia Pacific Conference on Circuits and Systems (APCCAS)

Enabling Resource-Efficient AI via SW-SW Co-Design Aug, 2025
University of Illinois Chicago (UIC)

Enabling Resource-Efficient AI via SW-SW Co-Design Aug, 2025
Duke University

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays Oct, 2023
International Conference on Computer Aided Design (ICCAD)

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays Sept, 2023
Sungkyunkwan University (SKKU) AI Colloquium

PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures Aug, 2023
International Symposium on Low Power Electronics and Design (ISLPED)

VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels Sept, 2022
Sungkyunkwan University (SKKU) AI Colloquium

VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures Mar, 2022
Design, Automation, and Test in Europe (DATE)

Teaching Experience

Teaching Assistant in Operating System
Sungkyunkwan University (SKKU), Suwon, Korea

- Instructor: Prof. Jong Hwan Ko

Mar, 2020

Student Guidance

Mentor for Undergraduate Research Project

Sept. 2025 - Present

Sungkyunkwan University (SKKU), Suwon, Korea

- Dynamic Computation Adjustment Techniques for IMC-based Efficient CNN Inference
- Yehyeon Kim, Seongjo Lee

Graduate Research Mentorship

Mar. 2023 - Mar. 2024

Sungkyunkwan University (SKKU), Suwon, Korea

- Kernel-Grouping Techniques for IMC Arrays [C7] - 🏆 [H12] Best Paper Award from KAIA (2023)
- Juhong Park

Mentor for Outstanding Undergraduate Research Program

Feb. 2022 - June. 2022

Sungkyunkwan University (SKKU), Suwon, Korea

- Remapping Method for Mitigating Cell Variation in IMC Arrays
- Minho Lee, Minseop Kim, Seonghoon Jeong, Jihoon Choi

Grants

Development of Collaborative Intelligent Agent Technologies for Human-AI Symbiotic Cooperation

July, 2025

Sungkyunkwan University (SKKU), Suwon, Korea

- Contributed to concept development and writing of the proposal
- Institute for Information & communications Technology Planning & Evaluation (IITP), Korea (Project # RS-2025-25442569)
- Amount awarded: ₩ 11.0 billion (\$ 8.15 million)

A Study on Collaborative Mapping and Training Techniques for Energy-Efficient In-Memory CNN Inference

July, 2024

Sungkyunkwan University (SKKU), Suwon, Korea

- Core idea development and proposal writing as a project leader
- National Research Foundation (NRF), Korea (Project # RS-2024-00409161)
- Amount awarded: ₩ 25 million (\$ 17.9 thousand)

Technical Reviewer Activities

IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)
Design Automation Conference (DAC)

Skills

Programming Python, PyTorch, VHDL, Verilog
Tools NeuroSim

References

Dr. Jong Hwan Ko (Ph.D. advisor)

Associate Professor, Dept. of Electrical and Computer Engineering,
Sungkyunkwan University (SKKU), Suwon, Korea
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Dr. Kang Eun Jeon

Postdoctoral Researcher, Kim Jaechul Graduate School of Artificial Intelligence,
Korea Advanced Institute of Science & Technology (KAIST), Seongnam, Korea
🏠 [Google Scholar](#) ✉️ kejeon@kaist.ac.kr